

```

MariaDB [(none)]> create database hotel;
Query OK, 1 row affected (0.029 sec)

MariaDB [(none)]> use hotel;
Database changed

MariaDB [hotel]> create table hotel (HotelNo int primary key, hotelName varchar(100) not null, city
varchar(100) not null);
Query OK, 0 rows affected (0.088 sec)

MariaDB [hotel]> create table Room (roomNo int not null, hotelNo int not null, type varchar(100)
not null, price decimal(10,2) not null, primary key (roomNo, hotelNo), foreign key (hotelNo)
references hotel (hotelNo));
Query OK, 0 rows affected (0.014 sec)

MariaDB [hotel]> create table Guest (guestNo int primary key, guestName varchar(255) not null,
guestAddress varchar(255));
Query OK, 0 rows affected (0.038 sec)

MariaDB [hotel]> create table Booking (hotelNo int not null, guestNo int not null, dateFrom date
not null, dateTo date, roomNo int not null, foreign key (hotelNo) references hotel (hotelNo),
foreign key (guestNo) references Guest (guestNo), foreign key (roomNo) references Room (roomNo),
primary key (hotelNo, guestNo, dateFrom));
Query OK, 0 rows affected (0.039 sec)

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-- Insertions

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MariaDB [hotel]> INSERT INTO Hotel VALUES (101, 'Grosvenor Hotel', 'London'), (102, 'Budget Inn',
'Mumbai'), (103, 'Sunrise Suites', 'Paris'), (104, 'Seaside Resort', 'Barcelona'), (105, 'Mountain
Lodge', 'Denver'), (106, 'City Center Hotel', 'New York'), (107, 'Grand Palace', 'Dubai'), (108,
'Lakeview Inn', 'Zurich'), (109, 'Desert Oasis', 'Cairo'), (110, 'Riverside Retreat', 'Bangkok');
Query OK, 10 rows affected (0.037 sec)
Records: 10 Duplicates: 0 Warnings: 0

MariaDB [hotel]> INSERT INTO Room VALUES (1, 101, 'Single', 45.00), (2, 102, 'Double', 15.00), (3,
103, 'Family', 75.00), (4, 104, 'Double', 55.00), (5, 105, 'Single', 65.00), (6, 106, 'Family',
90.00), (7, 107, 'Double', 80.00), (8, 108, 'Single', 60.00), (9, 109, 'Family', 95.00), (10, 110,
'Double', 70.00);
Query OK, 10 rows affected (0.030 sec)
Records: 10 Duplicates: 0 Warnings: 0

MariaDB [hotel]> INSERT INTO Guest VALUES (1001, 'John Smith', 'London'), (1002, 'Ravi Kumar',
'India'), (1003, 'Amina Diallo', 'Egypt'), (1004, 'Chen Wei', 'China'), (1005, 'Sofia Rossi',
'Italy'), (1006, 'Liam O'Connor', 'Ireland'), (1007, 'Priya Sharma', 'India'), (1008, 'Nina
Petrova', 'Russia'), (1009, 'Samuel Mensah', 'Ghana'), (1010, 'Isabel Garcia', 'Spain');
Query OK, 10 rows affected (0.028 sec)
Records: 10 Duplicates: 0 Warnings: 0

MariaDB [hotel]> INSERT INTO Booking VALUES (101, 1001, '2010-08-20', '2010-08-25', 1), (101, 1001,
'2027-07-01', '2027-07-05', 1), (102, 1002, '2011-11-05', '2011-11-10', 2), (103, 1003,
'2026-09-05', '2026-09-12', 3), (104, 1004, '2012-03-10', '2012-03-15', 4), (105, 1005,
'2026-11-20', '2026-11-25', 5), (106, 1006, '2027-01-05', '2027-01-10', 6), (107, 1007,
'2009-12-28', '2010-01-02', 7), (108, 1008, '2027-03-01', '2027-03-06', 8), (109, 1009,
'2027-04-10', '2027-04-14', 9), (110, 1010, '2027-05-15', '2027-05-20', 10);
Query OK, 11 rows affected (0.031 sec)
Records: 11 Duplicates: 0 Warnings: 0

```

Question 7.11

(a), (b), (c)

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MariaDB [hotel]> alter table Room Add constraint chk_type check(type in ('Single', 'Double',
'Family')), Add constraint chk_price check (price between 10.00 and 100.00), Add constraint
chk_roomNo check (roomNo between 1 and 100);
Query OK, 10 rows affected (0.066 sec)
Records: 10 Duplicates: 0 Warnings: 0

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(d)

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MariaDB [hotel]> alter table Booking Add constraint chk_dateOrder check (dateFrom < dateTo);
Query OK, 11 rows affected (0.062 sec)
Records: 11 Duplicates: 0 Warnings: 0

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MariaDB [hotel]> DELIMITER //
MariaDB [hotel]>
MariaDB [hotel]> CREATE TRIGGER before_booking_insert
-> BEFORE INSERT ON Booking

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-> FOR EACH ROW
-> BEGIN
->     IF NEW.dateFrom <= CURDATE()
->     OR NEW.dateTo <= CURDATE() THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Booking dates must be in the future';
->     END IF;
->
->     IF NEW.dateFrom >= NEW.dateTo THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'dateFrom must be before dateTo';
->     END IF;
-> END//

```

Query OK, 0 rows affected (0.015 sec)

(e)

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MariaDB [hotel]> DELIMITER //
MariaDB [hotel]>
MariaDB [hotel]> CREATE TRIGGER trg_no_room_double_booking
-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF EXISTS (
->         SELECT 1
->         FROM Booking b
->         WHERE b.roomNo = NEW.roomNo
->             AND NEW.dateFrom <= b.dateTo
->             AND NEW.dateTo >= b.dateFrom
->     ) THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Room is already booked for these dates';
->     END IF;
-> END//

```

Query OK, 0 rows affected (0.070 sec)

(f)

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MariaDB [hotel]> DELIMITER //
MariaDB [hotel]>
MariaDB [hotel]> CREATE TRIGGER trg_no_guest_overlap
-> BEFORE INSERT ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF EXISTS (
->         SELECT 1
->         FROM Booking b
->         WHERE b.guestNo = NEW.guestNo
->             AND NEW.dateFrom <= b.dateTo
->             AND NEW.dateTo >= b.dateFrom
->     ) THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Guest already has a booking during these dates';
->     END IF;
-> END//

```

Query OK, 0 rows affected (0.050 sec)

Question 7.12

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MariaDB [hotel]> CREATE TABLE BookingArchive LIKE Booking;
Query OK, 0 rows affected (0.031 sec)

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MariaDB [hotel]> insert into bookingArchive (Select * from Booking where dateFrom < '2013-01-01');
Query OK, 4 rows affected (0.030 sec)
Records: 4  Duplicates: 0  Warnings: 0

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MariaDB [hotel]> select * from bookingarchive order by dateFrom;

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hotelNo	guestNo	dateFrom	dateTo	roomNo
107	1007	2009-12-28	2010-01-02	7
101	1001	2010-08-20	2010-08-25	1
102	1002	2011-11-05	2011-11-10	2
104	1004	2012-03-10	2012-03-15	4

```

MariaDB [hotel]> Delete from Booking where dateFrom < '2013-01-01';
Query OK, 0 rows affected (0.030 sec)

```

```

MariaDB [hotel]> select * from booking order by dateFrom;
+-----+-----+-----+-----+-----+
| hotelNo | guestNo | dateFrom | dateTo | roomNo |
+-----+-----+-----+-----+-----+
| 103 | 1003 | 2026-09-05 | 2026-09-12 | 3 |
| 105 | 1005 | 2026-11-20 | 2026-11-25 | 5 |
| 106 | 1006 | 2027-01-05 | 2027-01-10 | 6 |
| 108 | 1008 | 2027-03-01 | 2027-03-06 | 8 |
| 109 | 1009 | 2027-04-10 | 2027-04-14 | 9 |
| 110 | 1010 | 2027-05-15 | 2027-05-20 | 10 |
| 101 | 1001 | 2027-07-01 | 2027-07-05 | 1 |
+-----+-----+-----+-----+-----+
7 rows in set (0.001 sec)

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Question 7.13

```

MariaDB [hotel]> CREATE VIEW CheapestHotels AS SELECT h.hotelNo, h.hotelName, h.city, r.price FROM
Hotel h JOIN Room r ON h.hotelNo = r.hotelNo WHERE r.price = (SELECT MIN(price) FROM Room);
Query OK, 0 rows affected (0.038 sec)

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MariaDB [hotel]> select * from cheapesthotels;
+-----+-----+-----+-----+
| hotelNo | hotelName | city | price |
+-----+-----+-----+-----+
| 102 | Budget Inn | Mumbai | 15.00 |
+-----+-----+-----+-----+
1 row in set (0.037 sec)

```

Question 7.14

```

MariaDB [hotel]> Create View brics_guests_view AS (Select * from Guest where guestAddress IN
('Brazil', 'Russia', 'India', 'China', 'South Africa', 'Egypt', 'Ethiopia', 'Indonesia', 'Iran',
'Saudi Arabia', 'United Arab Emirates'));
Query OK, 0 rows affected (0.034 sec)

```

```

MariaDB [hotel]> select * from brics_guests_view;
+-----+-----+-----+
| guestNo | guestName | guestAddress |
+-----+-----+-----+
| 1002 | Ravi Kumar | India |
| 1003 | Amina Diallo | Egypt |
| 1004 | Chen Wei | China |
| 1007 | Priya Sharma | India |
| 1008 | Nina Petrova | Russia |
+-----+-----+-----+
5 rows in set (0.003 sec)

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